Expanding synthesizable chemical space: benchmarking post-generation molecular modification

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Abstract

Synthesis-aware generative models produce drug candidates guaranteed to be synthesizable from commercial building blocks, but each is bounded by its training chemical space. PrexSyn, the current state of the art, reconstructs 94% of Enamine REAL molecules but only 28% of the broader ChEMBL bioactive distribution, leaving 72% of pharmacologically validated chemical space beyond its reach. We investigate whether post-generation modification methods expand this coverage when composed with the generative model. Four methods spanning rule-based and learned paradigms (CReM, mmpdb, JT-VAE, LibINVENT) are applied to PrexSyn's output across 100 ChEMBL-derived property specifications stratified by difficulty. Variants are evaluated on synthesizability, property conservation (substructural and physicochemical), drug-likeness, and chemical diversity, then ranked by marginal value over a repeated-sampling baseline. Complementarity analysis identifies optimal multi-method combinations, providing empirical guidance for extending synthesis-aware generation in drug discovery workflows.

Keywords: molecular generation, synthetic chemistry, drug discovery, machine learning

Research question and significance

Drug discovery frequently begins with a property wish list: a target molecular weight, lipophilicity, polar surface area, and related descriptors that define a pharmacological profile of interest. The practical challenge is to find synthesizable molecules that match it. Synthesis-aware generative models address this directly: given a property specification, they output molecules composed of purchasable building blocks joined via validated reactions, guaranteeing synthesizability by construction [1]. Several models operate in this paradigm, including SynNet, ChemProjector, SynFormer, and PrexSyn [2].

PrexSyn, the current state of the art, generates property-conditioned candidates within the Enamine REAL chemical space [3] at high throughput (∼0.26 s per target molecule), drawing from ∼480,000 in-stock building blocks and 115 reaction templates. It reconstructs 94% of molecules drawn from Enamine REAL but only 28% from ChEMBL [4], a curated database of bioactive compounds with demonstrated pharmacological activity. The remaining 72% of ChEMBL represents pharmacologically validated chemistry that lies beyond the model's reach.

This coverage gap is structural: every synthesis-aware generator is bounded by its building block library and reaction template set. Retraining on a larger chemical space shifts the boundary but does not eliminate it, and demands substantial data curation. Repeated sampling draws from the same constrained space, producing additional candidates but opening no new chemical territory. For discovery programs seeking novel scaffolds or unexplored regions of bioactive space, this constraint directly limits the generator's practical utility.

We investigate whether post-generation molecular modification expands the space of viable candidates when composed with the frontier synthesis-aware model. Fragment substitution, matched molecular pair transforms, latent-space editing, and scaffold-constrained decoration are established techniques that predate generative AI, operate outside Enamine REAL, and are available as open-source tools. We benchmark four such methods (two rule-based, two learned) on their ability to produce synthesizable, property-conserving candidates beyond what repeated PrexSyn sampling achieves. Results are stratified by how well PrexSyn performs alone (the baseline difficulty), revealing when modification adds the most marginal value. A complementarity analysis identifies which method combinations maximize total unique coverage, informing pipeline design and tool selection for both medicinal chemistry workflows and automated agent-driven design systems. This benchmark yields two core outputs: individual method rankings by marginal value, and optimal multi-method combinations that extend the generator's chemical reach without retraining.

Context and datasets

PrexSyn [2] generates synthesis pathways from ∼480,000 Enamine in-stock building blocks via 115 reaction templates, conditioned on molecular property specifications. We use it as a black-box generator: given a property profile, it returns synthesizable candidates within Enamine REAL.

ChEMBL [4] provides realistic property specifications grounded in demonstrated pharmacological activity: these molecules have been tested in real assays and recorded alongside their measured properties, making them representative of genuine drug discovery targets rather than hypothetical chemical space. We extract computed property profiles from sampled bioactive molecules: ECFP4 fingerprints (bit vectors encoding circular substructure features within a fixed bond radius, widely used for measuring molecular similarity) and physicochemical descriptors (molecular weight, lipophilicity, polar surface area, hydrogen bond donors and acceptors, and rotatable bond count). These profiles serve as input to PrexSyn. The molecules themselves are not optimization targets; we report structural similarity to the ChEMBL source only as a supplementary diagnostic (proximity to known bioactive space).

AiZynthFinder [5], trained on the USPTO reaction corpus [6], serves as our synthesizability gate. It performs retrosynthetic analysis: computational decomposition of each candidate into purchasable Enamine building blocks via known reaction steps. Enamine REAL represents a specific, commercially relevant subset of drug-like chemistry; molecules synthesizable through alternative routes, building block suppliers, or non-template reactions may not be detected.

The four modification methods draw on independent chemical knowledge: CReM [7] uses a context-constrained fragment database mined from ChEMBL; mmpdb [8] applies empirical matched molecular pair transformation rules; JT-VAE [9] encodes and decodes molecules via a learned continuous representation; LibINVENT [10] generates scaffold decorations via reinforcement learning (accessed through REINVENT4 [11]).

Proposed methods

For each property specification, PrexSyn generates an initial synthesizable candidate. Four modification methods then produce structural variants from this seed, and all variants pass through identical synthesizability and property-conservation filters. A repeated-sampling baseline (additional PrexSyn draws from the same specification) tests whether simply generating more candidates from the constrained space matches or exceeds the value of applying modification methods that operate beyond it.

CReM [7] modifies molecules by replacing chemical fragments (side chains, ring systems, functional groups) with alternatives drawn from a fragment database mined from real ChEMBL compounds, subject to chemical validity constraints. A tunable context radius controls the edit scale: radius 1 restricts changes to small R-group substitutions (modifications to a molecule's peripheral functional groups, e.g., swapping a methyl group for a chlorine atom, a change that can alter potency or metabolic stability); radius 2 permits medium-scale modifications such as ring substitutions; radius 3 enables scaffold-level replacements including entire ring-system swaps. This hierarchy mirrors how medicinal chemists optimize leads: iteratively modifying functional groups to improve potency, selectivity, or pharmacokinetics, starting conservatively and progressing to larger structural changes as needed. We benchmark CReM at context radii 1, 2, and 3 to characterize the edit-scale/quality tradeoff within a single method.

mmpdb [8] applies matched molecular pair (MMP) transformation rules: single-site substitution patterns extracted from pairs of real compounds that differ by exactly one structural change (e.g., methyl → chloro, phenyl → pyridyl). Each rule carries empirical property-change annotations derived from measured data, enabling selection of transformations likely to preserve the desired property profile. These are the most conservative edits in our benchmark: one atom or functional group at a time, with known property consequences. In medicinal chemistry, such transformations are a standard tool for systematically exploring the immediate chemical neighborhood of a lead compound (a promising early-stage candidate selected for further property optimization in a drug discovery program).

JT-VAE [9] encodes molecules into a continuous mathematical space (a learned latent representation) by decomposing each molecule's bonded structure into a tree of chemically meaningful substructures (a junction tree, where each node represents a ring system, bond, or atom cluster). JT-VAE generates variants by perturbing the seed molecule's latent vector and decoding the result, producing edits that span medium to global structural scales. This enables scaffold hopping: replacing an entire core structure while preserving the pharmacological profile, a strategy valued in drug discovery for accessing new chemical series and navigating intellectual property constraints.

LibINVENT [10, 11] preserves a molecular scaffold (the core ring system and connectivity pattern) extracted from PrexSyn's output while generating alternative R-group decorations (the functional groups attached to the scaffold) via reinforcement learning steered toward target properties. Where CReM and mmpdb edit at the fragment or single-site level, and JT-VAE operates across global structural scales, LibINVENT occupies a distinct niche: scaffold-constrained decoration. It systematically varies the peripheral chemistry that often drives selectivity and pharmacokinetic behavior, while holding the core structure fixed.

These four methods span two independent axes: the rule-based/learned axis (CReM and mmpdb apply deterministic rules from empirical databases; JT-VAE and LibINVENT use trained neural networks) and the local-to-global edit-scale axis (mmpdb → CReM → JT-VAE, with LibINVENT operating under scaffold constraints). Selection was informed by a systematic review of ∼50 candidate methods, prioritizing coverage across edit scales and paradigms. No prior work systematically benchmarks modification methods as post-generation complements to synthesis-aware generation. Beyond individual method rankings, a complementarity analysis identifies which method combinations maximize total unique coverage. These rankings and complementarity matrices provide the empirical basis for composing modification operators into multi-stage pipelines with synthesis filtering at every stage, a capability absent from existing tools and applicable to both medicinal chemistry workflows and automated agent-driven pipelines.

Data and preprocessing

We sample 100 molecules from ChEMBL with pre-stratified difficulty. PrexSyn is first run on a larger pool to compute baseline quality scores (ECFP4 Tanimoto similarity between the best candidate and the specification fingerprint, where Tanimoto measures the fraction of shared features between two binary fingerprints on a 0-to-1 scale). Approximately 25 molecules are then drawn from each of four quality bins (<0.5, 0.5–0.7, 0.7–0.85, 0.85–1.0), ensuring balanced representation across difficulty levels. This pre-stratification prevents easy specifications from dominating the benchmark and guarantees statistical power in every difficulty regime.

For each specification, PrexSyn generates 256 candidates; the highest-scoring candidate ($K = 1$) seeds all four modification methods, so differences in output reflect the modification method rather than seed selection. Top-K seeding ($K > 1$), which tests whether seed diversity compounds with method diversity, is a natural extension; we fix $K = 1$ here for clean causal attribution. Each method produces up to N variants; N is determined by a pilot saturation study (5 specs × 4 methods × 1,000 variants) that identifies the diversity plateau beyond which additional variants yield diminishing structural novelty. Expected cap: 100–500 variants per method per specification. Variants are deduplicated by canonical SMILES (a standardized text representation of molecular structure) before synthesizability checking via AiZynthFinder at search depth 1 (chosen for tractability; depth 2 is ∼5× slower), with depth-2 sensitivity analysis on a 10% subsample to quantify routes missed by shallower search. Estimated total load after deduplication: ∼80,000 unique molecules, processable in ∼8 core-hours.

No models are trained in this study; all methods operate as pre-trained or preconfigured tools. CReM's fragment database derives from ChEMBL, the same source as our property specifications. Because ChEMBL molecules serve only as property-profile sources and not as reconstruction targets, this overlap reflects realistic tool usage rather than information leakage.

Information flow diagram

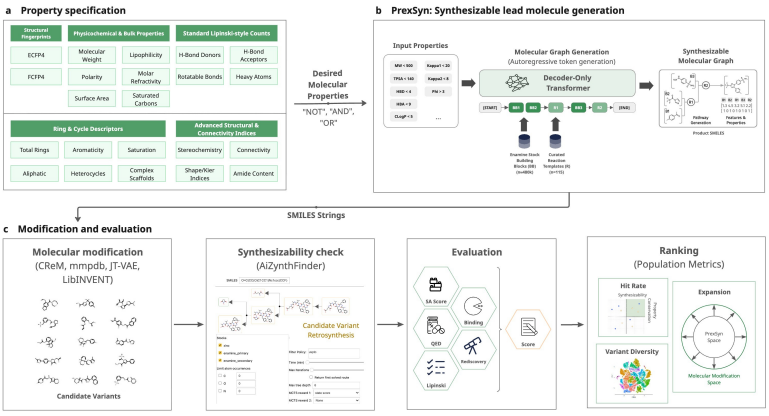

Fig. 1 Pipeline overview. (a) Target properties are defined via molecular descriptors and logical operators. (b) PrexSyn generates candidates from Enamine building blocks (n=480K) and reaction templates (n=115). (c) Four modification methods (CReM, mmpdb, JT-VAE, LibINVENT) diversify the top candidate per specification; variants are verified for synthesizability via AiZynthFinder and ranked by hit rate, chemical diversity, and expansion factor relative to a repeated-sampling baseline.

Experimental setup and baselines

Baseline. Repeated PrexSyn sampling: N additional draws from the same property specification, subject to identical evaluation. This tests whether adding a modification layer outperforms simply sampling more from the generative model. The pilot saturation study (Section "Data and Preprocessing") determines N at or beyond each method's diversity plateau, ensuring a fair comparison: neither method nor baseline is evaluated below its saturation point.

Hit classification. We classify a variant as a hit if it is (a) synthesizable (AiZynthFinder route found) and (b) property-conserving on both axes defined below. We treat thresholds as parameters, reporting results at Tanimoto $\geq$ 0.6, 0.7, and 0.85 (spanning lenient to stringent substructural conservation) and desirability $\geq$ 0.8, so conclusions are not artifacts of a single cutoff.

Property conservation is assessed on two complementary axes. *Substructural*: ECFP4 Tanimoto similarity between variant and specification fingerprint, measuring the fraction of shared circular substructure features. Recent benchmarks demonstrate that ECFP4-based similarity matches or exceeds pretrained molecular embeddings for property-relevant comparisons. *Physicochemical*: per-descriptor deltas (MW $\pm$50 Da, CLogP $\pm$1.0, TPSA $\pm$20 Å^2, HBD $\pm$1, HBA $\pm$2, rotatable bonds $\pm$2) mapped to 0–1 desirability scores via linear tolerance functions and aggregated by geometric mean. The geometric mean ensures that a single badly violated descriptor drives the aggregate score toward zero, penalizing unbalanced profiles more stringently than

an arithmetic mean would. These are the same descriptors PrexSyn conditions on, ensuring evaluation matches the generator's abstraction level. We report 3D shape and electrostatic similarity via ESPsim [12] as a supplementary diagnostic, not a hit-classification gate, because it introduces a dependency on the reference molecule's 3D conformer geometry (generated via the ETKDG conformer-generation algorithm) that couples evaluation to the specific ChEMBL molecule beyond its property values.

Primary metrics. (1) *Hit rate*: fraction of synthesizable variants meeting conservation thresholds; higher rates indicate more efficient use of computational budget. (2) *Expansion factor*: unique hits from modification divided by unique hits from baseline, the core measure of marginal value. An expansion factor >1 means the method produces viable candidates inaccessible to repeated sampling alone, expanding the chemical territory available to a discovery program without retraining the generator.

Secondary metrics. Quantitative Estimation of Drug-likeness (QED [13]), a 0–1 composite score aggregating desirability functions over eight molecular properties (a proxy for oral bioavailability and development tractability), reported as a distribution over each method's hits to assess whether modifications preserve or degrade drug-likeness. Chemical diversity among hits (1− mean pairwise ECFP4 Tanimoto), measuring whether a method produces structurally varied candidates or clusters around a single chemotype. Wall-clock compute cost per specification. Proximity to known bioactive space (Tanimoto to ChEMBL reference molecule, reported as a diagnostic distribution only).

Complementarity. Pairwise Jaccard overlap (shared hits divided by total unique hits across both methods) between method hit sets identifies redundant vs. complementary tools. We report the two- and three-method combinations that maximize total unique-hit coverage. Methods operating at different edit scales are likely to explore partially distinct chemical territory, but the degree of overlap is an empirical question: empirically measuring it determines whether multi-method pipelines are worth the added computational cost relative to using the single best-performing method alone. Low overlap indicates complementary methods; high overlap indicates redundancy. This analysis directly informs practical pipeline design: identifying the smallest set of methods that maximizes coverage guides resource allocation in both manual and automated discovery workflows.

Stratification. We report all metrics per baseline-quality bin (<0.5, 0.5–0.7, 0.7–0.85, 0.85–1.0) to reveal when post-generation modification adds the most marginal value. If modification helps most when the generator performs poorly, it serves as a rescue strategy for difficult specifications; if it helps uniformly, it is a general-purpose expansion tool. This stratification directly informs how modification methods should be deployed in practice.

Timeline to milestone 2

Week	Task	Lead	Notes / Bottleneck
Mar 9–16	PrexSyn environment setup and baseline inference		GPU access for PrexSyn
Mar 9–16	AiZynthFinder integration and validation		
Mar 9–16	Pilot saturation study (5 specs × 4 methods × 1K variants)		Determines variant cap N
Mar 16–23	Baseline generation (repeated PrexSyn sampling)		
Mar 16–23	Full variant generation (100 stratified specs × 4 methods)		AiZynthFinder throughput
Mar 23–30	Evaluation pipeline and complementarity analysis		
Mar 23–30	Figures and Milestone 2 draft		

Table 1 *

Bottleneck mitigation: Spec count reduced to 100 via pre-stratified sampling; variant count capped by pilot saturation; canonical-SMILES deduplication before AiZynthFinder; parallel execution on rented multi-core compute (∼80K molecules, under 1 hour on 16 cores).

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